index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Credit Card Security Demo</title>
    <link rel="stylesheet" href="style.css">
    <script src="https://cdnjs.cloudflare.com/ajax/libs/crypto-
js/4.1.1/crypto-js.min.js"></script>
</head>
<body>
    <div class="container">
        <h1>🔒 Credit Card Security Demo</h1>
        <p class="subtitle">Demonstrating Masking & AES Encryption</p>

        <div class="input-group">
            <label for="cardNumber">Credit Card Number</label>
            <input type="password" id="cardNumber" placeholder="Enter card
number" maxlength="19">
        </div>

        <div class="input-group">
            <label for="secretKey">Secret Key</label>
            <input type="password" id="secretKey" placeholder="Enter
encryption key">
        </div>

        <div class="button-group">
            <button class="btn-encrypt" onclick="maskAndEncrypt()">🔐 Mask
& Encrypt</button>
            <button class="btn-decrypt" onclick="decrypt()">🔓
Decrypt</button>
```

```html
        </div>

        <div id="outputSection" class="output-section hidden">
            <div class="output-group">
                <div class="output-label">Masked Card Number</div>
                <div class="output-value" id="maskedCard"></div>
            </div>
            <div class="output-group">
                <div class="output-label">Encrypted Data (AES)</div>
                <div class="output-value" id="encryptedData"></div>
            </div>
            <div class="output-group hidden" id="decryptedGroup">
                <div class="output-label">Decrypted Card Number</div>
                <div class="output-value" id="decryptedCard"></div>
            </div>
        </div>
    </div>

    <script src="script.js"></script>
</body>
</html>
```

Style.css

```css
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
    background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
    min-height: 100vh;
```

```
    display: flex;
    justify-content: center;
    align-items: center;
    padding: 20px;
}

.container {
    background: white;
    border-radius: 20px;
    box-shadow: 0 20px 60px rgba(0, 0, 0, 0.3);
    padding: 40px;
    max-width: 500px;
    width: 100%;
}

h1 {
    color: #333;
    text-align: center;
    margin-bottom: 10px;
    font-size: 28px;
}

.subtitle {
    text-align: center;
    color: #666;
    margin-bottom: 30px;
    font-size: 14px;
}

.input-group {
    margin-bottom: 20px;
}
```

```
label {
    display: block;
    color: #555;
    font-weight: 600;
    margin-bottom: 8px;
    font-size: 14px;
}

input {
    width: 100%;
    padding: 12px 15px;
    border: 2px solid #e0e0e0;
    border-radius: 8px;
    font-size: 16px;
    transition: border-color 0.3s;
}

input:focus {
    outline: none;
    border-color: #667eea;
}

.button-group {
    display: flex;
    gap: 10px;
    margin-bottom: 25px;
}

button {
    flex: 1;
    padding: 14px;
```

```
    border: none;
    border-radius: 8px;
    font-size: 16px;
    font-weight: 600;
    cursor: pointer;
    transition: all 0.3s;
}

.btn-encrypt {
    background: #667eea;
    color: white;
}

.btn-encrypt:hover {
    background: #5568d3;
    transform: translateY(-2px);
    box-shadow: 0 5px 15px rgba(102, 126, 234, 0.4);
}

.btn-decrypt {
    background: #48bb78;
    color: white;
}

.btn-decrypt:hover {
    background: #38a169;
    transform: translateY(-2px);
    box-shadow: 0 5px 15px rgba(72, 187, 120, 0.4);
}

.output-section {
    background: #f7fafc;
```

```
    border-radius: 10px;
    padding: 20px;
    margin-top: 20px;
}

.output-group {
    margin-bottom: 15px;
}

.output-group:last-child {
    margin-bottom: 0;
}

.output-label {
    font-size: 12px;
    color: #718096;
    font-weight: 600;
    margin-bottom: 5px;
    text-transform: uppercase;
    letter-spacing: 0.5px;
}

.output-value {
    background: white;
    padding: 12px;
    border-radius: 6px;
    font-family: 'Courier New', monospace;
    font-size: 14px;
    color: #2d3748;
    word-break: break-all;
    border: 1px solid #e2e8f0;
}
```

```
.hidden {
    display: none;
}

@media (max-width: 600px) {
    .container {
        padding: 30px 20px;
    }

    h1 {
        font-size: 24px;
    }

    .button-group {
        flex-direction: column;
    }
}
```

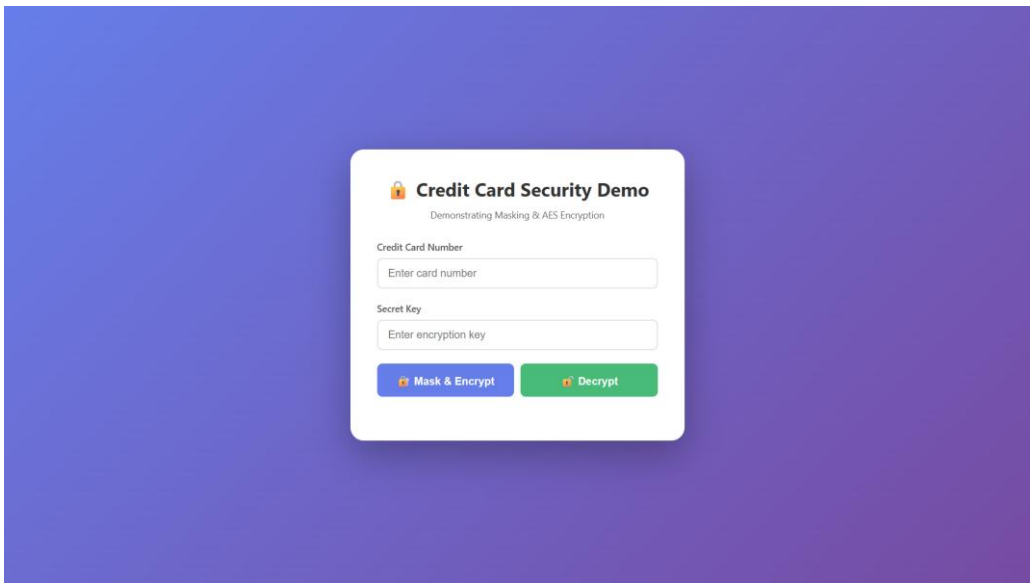
script.js

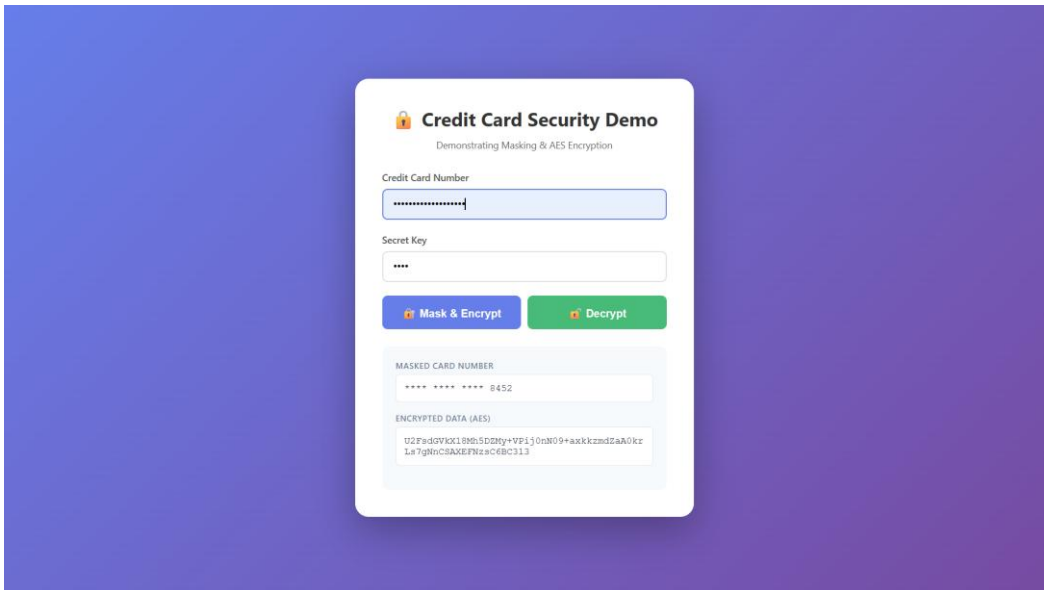
```
let encryptedCardData = null;

function maskAndEncrypt() {
    const cardNumber =
document.getElementById('cardNumber').value.replace(/\s/g, '');
    const secretKey = document.getElementById('secretKey').value;

    if (!cardNumber || cardNumber.length < 13) {
        alert('Please enter a valid credit card number (at least 13
digits)');
        return;
    }
```

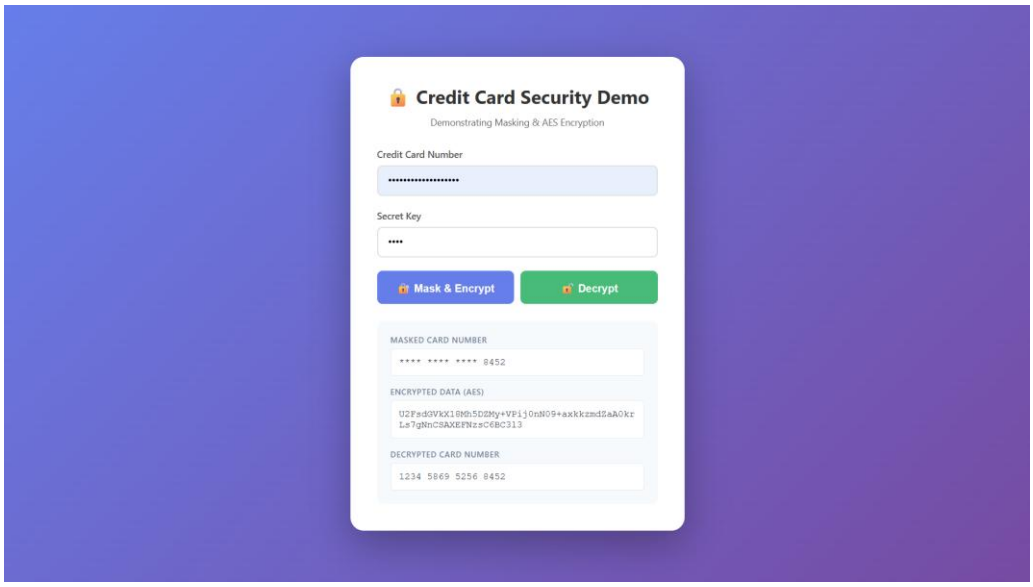
```
    if (!secretKey) {
        alert('Please enter a secret key for encryption');
        return;
    }

    // Mask the card number (show only last 4 digits)
    const lastFour = cardNumber.slice(-4);
    const masked = '**** **** **** ' + lastFour;

    // Encrypt the full card number using AES
    encryptedCardData = CryptoJS.AES.encrypt(cardNumber,
secretKey).toString();

    // Display results
    document.getElementById('maskedCard').textContent = masked;
    document.getElementById('encryptedData').textContent =
encryptedCardData;
    document.getElementById('outputSection').classList.remove('hidden');
    document.getElementById('decryptedGroup').classList.add('hidden');

    // Clear the card number input for security
    document.getElementById('cardNumber').value = '';
}
function decrypt() {
    const secretKey = document.getElementById('secretKey').value;

    if (!encryptedCardData) {
        alert('Please encrypt a card number first');
        return;
    }

    if (!secretKey) {
```

```
        alert('Please enter the secret key to decrypt');
        return;
    }

    try {
        // Decrypt the card number
        const bytes = CryptoJS.AES.decrypt(encryptedCardData, secretKey);
        const decrypted = bytes.toString(CryptoJS.enc.Utf8);

        if (!decrypted) {
            alert('Decryption failed. Please check your secret key.');
            return;
        }

        // Format the decrypted card number
        const formatted = decrypted.replace(/(\d{4})/g, '$1 ').trim();

        // Display decrypted result
        document.getElementById('decryptedCard').textContent = formatted;
        document.getElementById('decryptedGroup').classList.remove('hidden
');
    } catch (error) {
        alert('Decryption failed. Please check your secret key.');
    }
}

// Format card number input as user types
document.getElementById('cardNumber').addEventListener('input',
function(e) {
    let value = e.target.value.replace(/\s/g, '');
    let formattedValue = value.replace(/(\d{4})/g, '$1 ').trim();
    e.target.value = formattedValue;
});
```

Output:


Credit Card Security Demo
Demonstrating Masking & AES Encryption
Credit Card Number
Enter card number
Secret Key
Enter encryption key
Mask & Encrypt
Decrypt


Credit Card Security Demo
Demonstrating Masking & AES Encryption
Credit Card Number
Secret Key
Mask & Encrypt
Decrypt
MASKED CARD NUMBER
**** **** **** 8452
ENCRYPTED DATA (AES)
U2FsdGVkX18Mh5DZMy+VPij0nN09+axkkzmdZaA0kr
Ls7gNnCSAXEFNzsC6BC313


Credit Card Security Demo
Demonstrating Masking & AES Encryption
Credit Card Number
Secret Key
Mask & Encrypt
Decrypt
MASKED CARD NUMBER
**** **** **** 8452
ENCRYPTED DATA (AES)
U2FsdGVkX18Mh5DZMy+VPij0nN09+axkkzmdZaA0kr
Ls7gNnCSAXEFNzsC6BC313
DECRYPTED CARD NUMBER
1234 5869 5256 8452